

The Data Platform

STORE ROLES · EACH KIND OF DATA HAS EXACTLY ONE OWNER

- T Transactional truth**: service-owned PostgreSQL
the record others follow
built in the lab
- F Durable fact log**: JetStream, limits policy
append only, replayable
built in the lab
- W Workflow state**: Temporal persistence
runs in flight, not records
built in the lab
- R Derived read model**: projection plus KV cache
rebuildable, not the truth
built in the lab
- E Evidence documents**: repository behind adapter
write once, never edited
proposed · no V3 ADR

one change is committed once, then told to the others

OWNERSHIP · WHO MAY WRITE, AND WHO MAY NEVER LOOK

- A database per service**: own schema, own role, own migrations
ADR-052 · applied in the lab
built in the lab
- Shared instance by default**: isolation is logical, not physical
ADR-053 · accepted, not applied
proposed for V3
- The service mints the ID**: never the database, never a default
ADR-051 · a ULID, subject-safe
built in the lab
- One writer per record**: the owner writes, others are told
no second service updates it
built in the lab

a committed change becomes a published fact

THE WRITE PATH · ONE COMMIT, THEN ONE FACT, READ LEFT TO RIGHT

- Command accepted**: the domain rule is checked first
no rule lives in a handler
built in the lab
- Commit with an outbox**: state and its fact, one transaction
a fact cannot contradict the record
proposed · no V3 ADR
- Fact published**: on the owning service's stream
one publisher per subject family
built in the lab
- Projector writes both**: the projection and the cache
one handler, so they cannot drift
built in the lab

a reader never opens another service's database

THE READ PATH · CACHE, THEN PROJECTION, THEN A CONTRACT

- Cache read first**: NATS KV, keyed by context
the fast answer, not the truth
built in the lab
- A miss falls through**: to the projection table beneath
a cold cache is never an error
built in the lab
- The owner answers**: over a request-reply contract
the caller names no table
built in the lab
- Governed extraction**: reporting calls and bulk exports
the only route out of the platform
proposed · no V3 ADR

No foreign database read: X
not another service, not a report, not a script
a contract or nothing; a shared table is a hidden coupling
built in the lab · enforced by a database role per service

Keep history only where it must replay: !
otherwise current state alone is a plain record
replay when a rule, an audit or a rebuild reads the order
directional position as at 2026-09-04 · no V3 ADR of record

Read it downward. Five stores each hold one kind of data, one service owns each one, a single commit turns into a fact and then into a read model, and a reader is served by the cache, the projection or a contract. **Built today:** every solid tile — the database, the fact log, the projector, the cache and the read path. **Not shown:** subjects and streams, which are LB-V3-L3-05 ; where a store sits, which is LB-V3-L3-08 ; tables, columns and migrations, which are L4 .

WORDS ON THIS SHEET	
store a place data is kept, named by role	projector the code that applies a fact
fact a record that something happened	contract a published request the owner answers
fact Log the ordered, replayable history	KV NATS key-value store
projection a read model built from facts	ULID a sortable, subject-safe identifier
cache a fast copy of a projected answer	migration a versioned schema change
outbox the fact written with the state	ADR architecture decision record

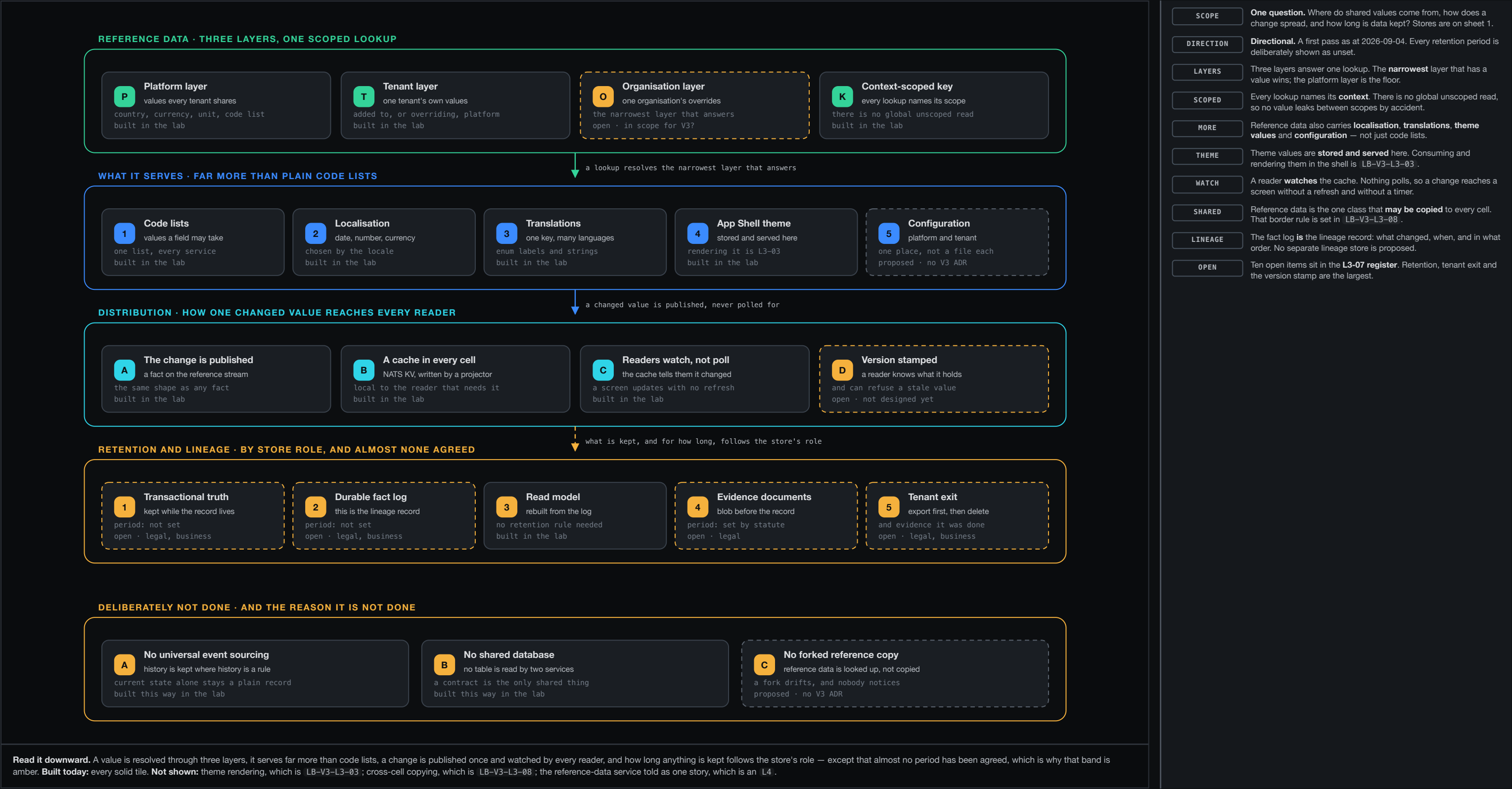
- **Cyan boundary** — the store roles being distinguished
- **Solid tile** — built and running in the lab today
- **Red panel** — a path that is refused outright
- **Arrow head** — the direction of the relationship

-  **Blue boundary** — the ordered path a write takes
-  **Dashed amber tile** — open, a named owner must confirm
-  **Blue line** — a committed change entering the write path

-  **Amber boundary** — the ordered path a read takes
-  **Amber panel** — a rule that governs the whole sheet
-  **Amber dashed line** — a written result reaching the read path

Reference Data, Distribution and Lifecycle

Where shared values come from, how one changed value reaches every reader without polling, and how long each kind of data is kept. Reference data is more than code lists: it also carries localisation, translation, App Shell theme values and platform configuration. **Maturity is marked per element**, and every retention period is deliberately shown as **not set**. **This half is directional** as at 4 September 2026 and no V3 decision of record exists for retention or lifecycle. Rendering the theme belongs to LB-V3-L3-03; what may be copied to another cell belongs to LB-V3-L3-08.



SCOPE	One question. Where do shared values come from, how does a change spread, and how long is data kept? Stores are on sheet 1.
DIRECTION	Directional. A first pass as at 2026-09-04. Every retention period is deliberately shown as unset.
LAYERS	Three layers answer one lookup. The narrowest layer that has a value wins; the platform layer is the floor.
SCOPED	Every lookup names its context . There is no global unscoped read, so no value leaks between scopes by accident.
MORE	Reference data also carries localisation, translations, theme values and configuration — not just code lists.
THEME	Theme values are stored and served here. Consuming and rendering them in the shell is LB-V3-L3-03 .
WATCH	A reader watches the cache. Nothing polls, so a change reaches a screen without a refresh and without a timer.
SHARED	Reference data is the one class that may be copied to every cell. That border rule is set in LB-V3-L3-08 .
LINEAGE	The fact log is the lineage record: what changed, when, and in what order. No separate lineage store is proposed.
OPEN	Ten open items sit in the L3-07 register . Retention, tenant exit and the version stamp are the largest.

LEGEND

Green boundary

the layers a lookup resolves through

Solid tile

built and running in the lab today

Blue line

a changed value entering distribution

Blue boundary

what reference data is asked for

Dashed grey tile

proposed for V3, not built

Amber dashed line

a store role imposing a lifecycle

Cyan boundary

how a change is distributed

Dashed amber tile

open, a named owner must confirm

Arrow head

the direction of the relationship

Amber boundary

unsettled, or deliberately refused

Green line

a lookup resolving to one answer

Line label

names the relationship, never bare "uses"

WORDS ON THIS SHEET

reference data shared values many services read

code list the values one field may take

localisation values shaped for one place

translation one key rendered per language

context the business-unit scope of a lookup

theme value a colour, size or font token

watch being told of a change, not asking

lineage the ordered record of what changed

retention how long something is kept

cell one region's complete runtime

KV NATS key-value store

ADR architecture decision record